Journal of Personality and Social Psychology
1976, Vol. 34, No. 4, 605–614

Communication Modality as a Determinant of Message Persuasiveness and Message Comprehensibility

Shelly Chaiken and Alice H. Eagly
University of Massachusetts—Amherst

Subjects were exposed to either an easy- or difficult-to-understand persuasive message presented via the written, audiotaped, or videotaped modality. With difficult messages, both persuasion and comprehension of persuasive material were greater when the message was written as compared with videotaped and audiotaped. With easy messages, persuasion was greatest when the message was videotaped, moderate when audiotaped, and least when written, but comprehension was equivalent regardless of modality. The persuasion and comprehension findings, as well as results on other variables, were discussed within a framework that considered the effect of communication modality on the capacity for effective reception of information as well as on the tendency to yield to information. The communicator's nonverbal expressions of confidence, manipulated within audiotaped and videotaped conditions, did not affect opinions.

A relatively unexplored question in attitude change research concerns the effects of the medium or channel of transmission upon the effectiveness of persuasive communications. Most previous research indicates that live or videotaped messages induce greater opinion change than oral (i.e., audiotaped) messages (Frandsen, 1963), which, in turn, effect greater change than written messages (Cantril & Allport, 1935; Haugh, 1952; Knower, 1935, 1936; Wilke, 1934). However, a few studies show either no differential persuasiveness as a function of modality or greater opinion change with written communications (McGinnies, 1965; Tannenbaum & Kerrick, 1954; Worchel, Andreoli, & Eason, 1975). In contrast to these persuasion findings, research on comprehension indicates that written presentation of material generally results in greater comprehension than does audiotaped or videotaped presentation (Harwood, 1951; Haugh, 1952; Wilson, 1974; Young, 1953), although this advantage may be manifest only with relatively complex material (Beighley, 1952; Carver, 1935; Harwood, 1951; Westover, 1958).

This greater capacity for the written modality to convey information seems somewhat paradoxical in conjunction with the majority of persuasion findings when one considers the assumption that good reception of a message generally facilitates opinion change (e.g., Hovland, Janis, & Kelley, 1953; McGuire, 1968, 1972), an assumption that has received empirical support (e.g., Eagly, 1974; Haaland & Venkatesan, 1968; Janis & Rife, 1959; McGuire, 1957). Although this relationship is generally considered to be the result of comprehension influencing the amount of information that the message recipient obtains in support of the message's conclusion, it may also be a product of the negative affect generated by the recipient's attempt to comprehend material that is not easily understood (Eagly, 1974).

This tendency for comprehension of message content to enhance persuasion suggests that the lack of consistency across previous modality studies may have been due to failures to control for message comprehensibility. Because the written mode's superiority in conveying information is manifest primarily with

This paper is based on the first author's master's thesis, which was directed by the second author. A preliminary version of the paper was presented at the meetings of the Eastern Psychological Association, New York, April 1974. The authors would like to thank Burt Franzman for serving as the communicator in the audiotape and videotape conditions of this study. This research received support from National Science Foundation Grant GS-40167.

Requests for reprints should be sent to Alice H. Eagly, Department of Psychology, University of Massachusetts, Amherst, Massachusetts 01002.

inherently difficult-to-understand material, modality differences in persuasiveness may favor the written mode with difficult material, whereas the more usual advantage of the audiotaped and videotaped (or live) modalities may be limited to easy material.

To examine this idea, subjects were presented with either an easy- or difficult-to-understand message in either the written, audiotaped, or videotaped mode. Modality differences in comprehension favoring written over audiotaped and videotaped messages were predicted only for difficult material. Therefore, in difficult conditions, written messages were expected to be more persuasive than audiotaped and videotaped messages. With easy material, which was expected to be comprehended equally well across modalities, it was anticipated that persuasion would be greater in videotaped compared with audiotaped and in audiotaped compared with written conditions.

Underlying these predictions are the assumptions that (a) people more readily *yield* to (McGuire, 1968, 1972) or accept information transmitted in a videotaped compared with audiotaped compared with written mode, and (b) with difficult messages, these yielding differences are outweighed by the written mode's persuasive advantage, which stems from its superiority in conveying difficult material. The assumption of modality differences in yielding is supported by the majority of opinion change findings cited earlier and by research suggesting that people are more critical of and perceive as less valid material that is written as opposed to material presented via audiotape or videotape (e.g., Carver, 1935; Maier & Thurber, 1968). Further, the study explored two potential explanations for modality differences in yielding. Both explanations, one involving distraction and the other dealing with perception of the communicator, are based on possible functions served by nonverbal cues, which are present in both audiotaped and videotaped modes but absent in the written modality.[1]

The distraction explanation suggests that vocal and visual cues may distract the message recipient from message content. To account for the presumed modality differences in yielding, this distraction must lessen the recipient's critical attitude toward the message, probably by reducing the ability to counterargue (Keating & Brock, 1974; Osterhouse & Brock, 1970), yet not be so extreme that it interferes with message comprehension (Baron, Baron, & Miller, 1973; Haaland & Venkatesan, 1968; Insko, Turnbull, & Yandell, 1974). To examine the role of distraction, both counterarguing and perceived distraction were assessed.

A second explanation for modality differences in yielding is that nonverbal cues in the videotaped and audiotaped modalities may affect perception of the communicator and, in particular, may increase the salience of aspects of the communicator's image that facilitate persuasion (cf. Williams, 1975). To explore this idea, the study assessed subjects' perceptions of the communicator.[2] In addition, because nonverbally expressed confidence is one aspect of the communicator's image that has been shown to enhance persuasion (London, 1973), this variable was manipulated within audiotaped and videotaped conditions. If typical modality differences in yielding (video > audio > written) are a product of the communicator's image, these differences should be maximized with confidently presented messages and minimized or reversed with nonconfidently presented messages.

In summary, the study examined whether differences in message comprehensibility account for modality effects in persuasion and, as a secondary focus, explored possible determinants of presumed modality differences in yielding.

---

[1] The distraction and communicator-image explanations for yielding differences were deemed reasonable from the standpoint of examining modality effects in a laboratory context. It seems likely, however, that to explain modality effects in other contexts, additional mechanisms, which are perhaps more salient outside of the laboratory, may need to be considered. For example, differing degrees of credibility may be attributed to modalities as a result of their history of use in a particular culture.

[2] This dependent variable, as well as the perceived distraction measure mentioned above, may reflect differences in message comprehensibility in addition to modality effects in yielding.

METHOD

Overview

Subjects indicated their opinions and other responses after receiving an easy- or difficult-to-understand persuasive message via written, audiotaped, or videotaped presentation. Within audiotape and videotape conditions, the communicator delivered the message confidently or nonconfidently.

Subjects

A total of 321 University of Massachusetts psychology students (154 males, 167 females) participated and received extra credit toward their course grades. Seven subjects were eliminated because they either suspected an influence attempt (1) or doubted the authenticity of the cover story (6); 15 more were randomly discarded in order to equalize cell frequencies.

Procedure

Participating in numbers ranging from 2 to 7 persons, subjects were told that the psychology department was helping a New England law school evaluate the ability of its law students to argue legal cases. The experimenter explained that the law school had recently implemented a training program and added that if the program were evaluated positively, the law school wanted its students to serve as student lawyers in community legal clinics. The experimenter then summarized what the subjects' participation entailed: reading background information about a case, reading (or listening to, or viewing) a randomly selected law student's "case discussion," giving their own opinions on the case, and responding to a "case evaluation" questionnaire.

The experimenter then allowed subjects 5 minutes to read handouts that provided background facts concerning the (fictional) "Victoria Company Case," a company–union dispute over the management's failure to pay a traditionally awarded Christmas bonus to workers. The handout gave both union and company positions in the case (i.e., that workers should vs. should not be compensated for the lost bonus) and concluded by stating that because an agreement had not been reached, an independent arbiter had been called in.

Next, the experimenter distributed written transcripts or played an audiotape or videotape of the persuasive message ("the law student's case discussion"). Afterwards, subjects indicated their agreement with the message. Then they received a "case evaluation" questionnaire containing the remaining dependent measures (see below). Finally, subjects were debriefed.

Control subjects ($n = 47$) were treated essentially the same but gave their opinions without being exposed to a persuasive communication.

Independent Variables

Message difficulty. The topic employed (a legal dispute) afforded the opportunity to manipulate message difficulty without rendering either the easy or difficult message implausible. Both versions were approximately 955 words long, advocated the company's position in the case, and contained three lengthy arguments supporting that position. The easy version was composed of sentences approximately 20 words in length with one or two clauses per sentence. The difficult version was composed of sentences approximately 30 words in length with three or more clauses per sentence. In addition, the difficult version differed from the easy one in terms of its more sophisticated vocabulary. For example, in the difficult version, the words "apportioned," "subducted," and "mollify" appeared in place of the words "distributed," "withdrew," and "change," which were used in the easy version. To illustrate further, one passage from the easy version read as follows:

> Of particular importance is the fact that the company agreed to make the wage increase granted in October of 1972 retroactive to September 1972. Making the wage increase retroactive cost the company over $9,300.

The corresponding passage in the difficult version was:

> One particular benefit which should be underscored is that of retroactivity to September, 1972 of the wage accretion granted in October of the same year, a benefit which the company financed and the cost of which exceeded $9,300.

Aside from differences in sentence structure and level of vocabulary used, the content of both versions was essentially identical.[3]

Confidence. An amateur actor was coached to deliver each of the above versions of the message using either confident or nonconfident paralanguage (i.e., tone of voice) and body language (i.e., gestures, posture).

Modality. The actor's final performances were videotaped and used in videotape conditions, whereas the audio portions of the tapes were played in audiotape conditions. These messages lasted approximately 7 minutes, and subjects in written conditions were allotted 7 minutes to read through written versions.

---

[3] The difficulty manipulation should not be considered a comprehensibility manipulation because message comprehension should be a joint product of both message difficulty and modality. An independent assessment of the success of the difficulty manipulation is virtually impossible because all likely checks should also be affected by modality. Thus, while an obvious overall check on the manipulation's effectiveness would be a finding of decreased comprehension given difficult as compared with easy messages, we expect that this effect should primarily result from lowered comprehension in audiotape and videotape (but not written) conditions.

Dependent Variables

Opinions. Subjects indicated their agreement with the statement, "The Victoria Company *should be required* to pay compensation to their union employees for the lost Christmas bonus," on a 15-point scale anchored by "definitely agree" and "definitely disagree." Disagreement with this statement implies agreement with the position advocated in the message.

Comprehension. The following four measures of message comprehension were employed: (a) a self-report of how difficult it was to understand the law student's discussion (15-point scale); (b) recall, assessed by having subjects write each of the three arguments that the law student had used to support his position (an answer was considered correct if, in the opinion of two independent raters, $r = .89$, it adequately summarized one of the arguments in the message); (c) recognition, assessed by a five-item, multiple-choice test on details of the legal dispute; (d) recall, assessed by a six-item, short-answer quiz concerning additional details (scored for correctness by two raters, $r = .91$). Also, subjects wrote down the overall position taken by the law student. All subjects demonstrated satisfactory recall on this measure.

Perception of the source. Subjects rated the source (law student) on 15-point bipolar evaluative scales. Positive poles were competent, warm, intelligent, approachable, interesting, generous, sincere, friendly, unbiased, modest, well read, good sense of humor, persuasive, and confident.

Other measures. Subjects indicated, on 15-point scales, how distracted they felt from the content of the case discussion, as well as how much effort they had expended and how pleasant they had found the experience of reading (or listening to, or viewing) the case discussion. Again on 15-point scales, subjects rated the extent to which they thought that the law student's language was appropriate for use in a community legal clinic and the extent to which they believed that he was an appropriate person to work in such a clinic. To assess counterarguing, subjects listed their thoughts and ideas concerning the consequences of the position advocated in the message. A statement was scored as a counterargument if, in the opinion of two raters, $r = .94$, it indicated a negative consequence of the position advocated or constituted an attack on the message's validity (cf. Brock, 1967; Osterhouse & Brock, 1970).

Subjects also wrote down their interpretations of the study. These responses were coded for suspicion.

Results

In addition to performing overall significance tests through analyses of variance on the dependent measures, the major hypotheses were explored by simple effects tests and planned comparisons implied by the predictions detailed earlier. The four comprehension measures were treated by multivariate analysis of variance. Since subjects' sex had no significant effect on opinions and few significant effects elsewhere, all reported analyses were performed ignoring this variable.

Manipulation Checks

Although subjects in confident conditions perceived the source as more confident than subjects in nonconfident conditions ($p < .001$), confidence did not significantly affect opinions and showed few effects on other measures. Thus, all subsequent analyses ignored this variable. The multivariate analysis of the comprehension measures indicated significantly poorer message comprehension for subjects in difficult compared with easy conditions, $F(4, 243) = 7.19$, $p < .001$. All four univariate components of the multivariate analysis also exhibited this effect ($p < .005$ or smaller).

Opinions

All treatment groups differed significantly from the control group in the direction of greater agreement with the message ($p < .05$ or smaller by Dunnett's test; cf. Myers, 1972). Opinion change scores (see Table 1) were formed by subtracting from each subject's opinion the mean of the control group ($M = 8.27$).

An analysis on these scores yielded a Modality × Message Difficulty interaction, $F(2, 246) = 7.89$, $p < .001$. As predicted, when easy messages were received, opinion change was greater in videotape over audiotape over written conditions, simple $F(2, 246) = 3.56$, $p < .05$, although planned comparisons revealed that only the videotape versus written difference was significant ($p < .01$). The modality effect was also significant within difficult conditions, simple $F(2, 246) = 6.41$, $p < .005$, and a planned comparison supported the prediction of significantly greater opinion change in written conditions relative to videotape and audiotape conditions combined ($p < .01$).[4]

[4] A comparison of the written-easy and written-difficult cells reveals that relatively (though not significantly) greater opinion change occurred when subjects received difficult rather than easy written

TABLE 1

DEPENDENT VARIABLES AS A FUNCTION OF MODALITY AND MESSAGE DIFFICULTY

	Message difficulty					
	Easy			Difficult		
Variable	Written	Audio-tape	Video-tape	Written	Audio-tape	Video-tape
Opinion change	2.94	3.75	4.78	4.73	2.32	3.02
Perceived difficulty	4.76	4.21	4.83	5.31	7.50	7.43
No. persuasive arguments recalled	2.45	2.21	2.17	2.29	1.74	1.67
No. short-answer items correct	4.57	3.93	4.45	4.21	3.71	3.36
No. multiple choice items correct	3.55	3.05	3.24	3.10	2.81	2.64
Perceived pleasantness	9.00	9.07	9.50	8.38	6.81	7.38
Perceived distraction	4.67	5.69	6.98	6.79	9.21	8.24
Appropriateness of language	11.33	10.83	10.19	9.60	7.62	7.45
Appropriateness of communicator	9.67	10.07	9.76	9.83	7.36	8.21

Note. Higher numbers indicate greater opinion change and message comprehension; greater amounts of perceived difficulty, pleasantness, and distraction; and a greater tendency to judge the communicator as well as his language appropriate for a legal clinic. (Cell $n = 42$.)

Comprehension and Pleasantness

In addition to the significant manipulation check on comprehension, the multivariate analysis yielded a significant modality effect, $F(8, 486) = 2.23$, $p < .05$. This effect was significant on three univariate components (arguments recalled, multiple-choice, and short-answer scores, $p < .05$ or smaller) and marginal on one (self-reported difficulty, $p = .13$). The modality effect indicates that, overall, subjects in written conditions comprehended more of the message than subjects in either videotape or audiotape conditions.

The Modality × Message Difficulty interaction was marginally significant in the multivariate analysis, $F(8, 486) = 1.81$, $p = .07$. It was significant on one univariate component, self-reported difficulty ($p < .05$) and marginal on a second, number of short-answer items correct ($p = .13$). The patterning of the interaction on these variables, as well as a similar but nonsignificant patterning on the remaining components, is consistent with the prediction that the superior comprehensibility of written messages would be manifest only with difficult-to-understand messages (see Table 1). Thus, within easy conditions the simple modality effect was nonsignificant on all four comprehension measures and indicates that comprehension was essentially identical regardless of mode of presentation. Within difficult conditions, however, the modality effect was significant on three measures: perceived difficulty ($p < .01$), arguments recalled ($p < .005$), and short-answer items correct ($p < .05$). Further, planned comparisons revealed that given difficult messages, subjects in written conditions, relative to subjects in videotape and audiotape conditions combined, reported less difficulty ($p < .01$), recalled more persuasive arguments ($p < .01$), and correctly answered more short-answer items ($p < .05$).

Regarding the pleasantness ratings (see Table 1), subjects receiving easy messages perceived their experiences as more pleasant than those receiving difficult messages, $F(1, 246) = 21.58$, $p < .001$. The Modality × Message Difficulty interaction was only marginally significant on this variable, $F(2, 246) = 2.24$, $p = .11$. However, paralleling the comprehension findings, the simple effect of modality within easy conditions was nonsignificant, $F < 1.0$, whereas the simple effect within difficult conditions was significant ($p < .05$). Thus, given easy messages, perceived pleasantness was essentially identical, regardless of modality, but given difficult mes-

messages. Obviously, the two messages did represent two unique communications even though the difficulty manipulation was designed to create messages differing solely in ease of comprehension. For this reason and because the substantive empirical question concerned the relative persuasive effectiveness of the various modalities as a function of message difficulty, the Modality × Message Difficulty interaction was explored by examining differences among modality conditions within levels of message difficulty.

sages, pleasantness was greater in written conditions compared with audiotape and videotape conditions combined ($p < .05$, planned comparison).

Other Dependent Variables

Analyses of subjects' appropriateness ratings (see Table 1) revealed that subjects receiving easy, compared with difficult, messages rated both the communicator and his language as more appropriate for a legal clinic setting (ps $< .001$). Also, subjects in written conditions, relative to subjects in audiotape and videotape conditions, viewed the communicator's language as significantly more appropriate ($p < .005$) and viewed the communicator as slightly more appropriate for clinic work ($p = .07$). Finally, the Modality × Message Difficulty interaction on the communicator's appropriateness for clinic work ($p < .005$) indicated that given easy messages, appropriateness ratings did not vary with modality, whereas given difficult messages, subjects in written conditions viewed the communicator as more appropriate for clinic work than did subjects in audiotape and videotape conditions.

On the distraction ratings (see Table 1), the modality effect ($p < .001$) supported the expectation that subjects in written conditions would report less distraction from message content than would subjects in audiotape and videotape conditions. In addition, subjects receiving difficult, compared with easy, messages reported greater distraction ($p < .001$).

A factor analysis of the source ratings yielded two rotated factors, labeled "professionalism/expertise" (e.g., confident, intelligent, competent) and "personal attractiveness/warmth" (e.g., generous, approachable, friendly), which accounted for 24.70% and 17.40% of the variance, respectively. Factor scores, obtained for each subject on each of these factors, were treated by analysis of variance. Subjects receiving easy messages viewed the communicator as more professional and expert ($p < .05$) and more personally attractive and warm ($p < .05$) than did subjects receiving difficult messages. Also, subjects in written conditions perceived the source as more professional and expert than did subjects in auditotape and videotape conditions ($p < .001$). Finally, subjects in audiotape conditions viewed the communicator as less attractive and warm than did subjects in written and videotape conditions ($p < .001$).

Analyses on perceived effort and counterarguing yielded no significant effects.

Discussion

Our analysis of the impact of communication modality on persuasion suggested taking into account the effect of modality on reception of information as well as on the tendency to yield to information that is received. The written mode's superiority in transmitting information was expected to be manifest only with difficult material. Thus, the increment in persuasiveness stemming from superior comprehension in the written mode was expected to appear only in the difficult message condition of the experiment. It was therefore predicted that messages conveying difficult content would be more persuasive if written rather than videotaped or audiotaped. In contrast, because easy messages should minimize modality differences in comprehension while maintaining differences in yielding, they were predicted to be more persuasive if videotaped rather than audiotaped and if audiotaped rather than written. The results on opinion change supported these predictions.

Comprehension and Pleasantness as Mediators of Persuasion

Additional support for our analysis can be seen in the comprehension findings, which were consistent with its presumed mediating role in relationship to opinion change. Although the interaction between modality and message difficulty fell short of significance ($p = .07$) in the multivariate analysis, simple effects tests and planned comparisons showed that for difficult messages, comprehension was greater in the written than in the videotaped and audiotaped modalities, whereas the modalities differed little for easy messages.

Since the comprehension indices presumably served to measure the reception mediator directly, the fact that we did not obtain as strong an interaction effect on message comprehension as we did on opinions deserves comment. As McGuire (1969) has noted, a recurrent "paradox" in research guided by

learning theory and other approaches to attitude change is that an independent variable, while showing the predicted relationship to the distal attitude change variable, exhibits weakly, if at all, the predicted relationship to the proximal learning or information-processing variable, even though the hypothesized proximal effect served as a basis for predicting the distal one. This paradox becomes somewhat understandable, however, when one considers that, like most research in this area, the present study placed some reliance on verbal memory (i.e., message recall and recognition) in order to document the mediational role of message reception. As has been argued (e.g., Eagly, 1974), memory provides only a weak indication of reception because verbal memory reflects not only reception, or encoding processes, but also storage and decoding processes. In this regard, it is interesting to note that in the multivariate analysis, the univariate component on which the Modality × Message Difficulty effect appeared most strongly was subjects' self-reported difficulty in understanding the communication. Perhaps this "subjective" measure more closely reflected the quality of subjects' reception experiences than did our more "objective" recall and recognition measures.

Subjects' ratings of the pleasantness of receiving the message paralleled the comprehension findings: With difficult messages the written modality tended to engender a more pleasant experience than the audiotape and videotape modalities, but with easy messages the modalities differed little. This parallelism supports Eagly's (1974) argument that the decrement in opinion change that follows from poor comprehension may stem from a negative affective reaction as well as from the lesser amount of information received. To compare these two mediators, correlational analyses were undertaken. These analyses revealed that information received (as estimated by a retention index formed by summing over each subject's three message comprehension scores) and pleasantness each correlated more highly with opinions ($r_{\text{Retention}} = .33$; $r_{\text{Pleasantness}} = .34$) than with each other ($r = .21$). Further, a multiple regression analysis, including both information received and pleasantness as predictors, revealed that each variable contributed significantly in predicting opinion change, $\beta_{\text{Retention}} = .27$, $F(1, 249) = 21.23$, $p < .001$; $\beta_{\text{Pleasantness}} = .29$, $F(1, 249) = 24.11$, $p < .001$. These findings, as well as the results of analyses of covariance controlling opinion change separately for comprehension and pleasantness, suggested that the amount of information received and perceived pleasantness both contributed to the opinion change effects and did so with about equal strength.

Additional support for the importance of affect in contributing to reception effects may be provided by the appropriateness ratings. Thus, in the difficult audiotape and videotape conditions, subjects judged both the communicator and his language as relatively inappropriate for a legal clinic setting. Perhaps these ratings, which correlated significantly with pleasantness (ps $< .001$), reflected annoyance with a communicator who was perceived as talking in an unnecessarily complicated manner. However, another explanation of these ratings might be that people expect oral presentations to be less complex than written ones—perhaps because communicators typically do adjust the difficulty of their material to the limitations imposed by channels of transmission. When expectancies are violated, as they probably are by audiotaped or videotaped presentations conveying very complex material, people consider these presentations inappropriate.

The present findings extend the generality of the Eagly (1974) research because our difficulty manipulation, accomplished through variations in vocabulary and sentence structure, is probably more representative of the ways in which differences in comprehension are created in more naturalistic situations than were the comprehensibility manipulations used in her earlier experiments (e.g., noise and distortion on an audiotaped message). It should be emphasized, however, that our difficulty manipulation was analogous to Eagly's comprehensibility manipulations only with respect to the audiotape and videotape conditions of the present experiment (see Footnote 3). In a Modality × Message Difficulty analysis of variance excluding written conditions, we obtained, like Eagly, a message

difficulty main effect on opinions ($p < .005$), as well as on all the comprehension measures (all ps $< .05$ or smaller) and on perceived pleasantness ($p < .001$). That the interaction between modality and message difficulty, but not a message difficulty main effect, obtained on opinions when all modality levels were considered is due, as noted earlier, to the fact that the difficulty manipulation had, as predicted, little impact on written subjects' abilities to comprehend the message and, thus, little impact on their opinions.

Modality Differences in Yielding

Although reception processes can acount for the lowered persuasiveness of difficult messages in the audiotaped and videotaped compared with written mode, modality differences in yielding must be invoked to explain the ordering obtained with easy messages (video > audio > written). Two possible explanations for these yielding differences were explored.

One explanation proposed that nonverbal cues, which decrease in number from videotaped to audiotaped to written modalities, diminish receivers' critical abilities. Supporting this interpretation, subjects in written conditions reported less distraction from message content than did subjects in audiotape conditions who, in turn, reported less distraction than did subjects in videotape conditions. However, because easy messages were rated less distracting than difficult ones, it seems that distraction is best viewed as a complex variable that may enhance yielding by reducing the ability to be critical but may also interfere with message reception (e.g., Insko et al., 1974).

Though the effect of modality on distraction provides some support for the distraction mechanism, it should be noted that the experimental variables failed to affect counterarguing, a variable which past research (e.g., Osterhouse & Brock, 1970) has implicated as a mediator of enhanced opinion change under distraction conditions. Yet, as Osterhouse and Brock argued, it may be necessary for a persuasive message to counter a well-anchored attitude in order for counterarguing to occur. In contrast, our message presented new information to subjects on an unfamiliar topic.

Our analysis also suggested that the increase in yielding as modality changes from written to audiotape to videotape could be a product of an enhancement of the communicator's image from the addition of nonverbal vocal and visual cues. However, the source ratings were not more favorable in videotape and audiotape relative to written conditions. The communicator image mechanism also failed to receive support from the manipulation of communicator confidence, which neither affected opinion change nor mediated the effect of modality on opinions.[5] In summary, with respect to exploring the determinants of modality differences in yielding, no support was obtained for the communicator image mechanism, whereas partial support was obtained for the distraction mechanism. Because the underlying reasons for modality differences in yielding remain unclear, future research might well address itself to a more thorough exploration of this issue.

Conclusions

The present research indicates that the effects of communication modality on persuasion can be usefully viewed within a framework that stresses the importance of effective reception of persuasive material. By observing the hypothesized comprehension effects, in addition to the predicted persuasion findings, we can make a relatively strong inference regarding the causal role of comprehension. In this regard, it is interesting to compare our findings with those of Miller, Maruyama, Beaber, and Valone (1976). With an audio message they obtained no effect of message "complexity," manipulated through changes in sentence structure similar to those used in our difficulty manipulation. While our findings suggest that their complex message should have been less persuasive than the simple one, it is not possible to interpret their finding confidently in terms of comprehension mediation

[5] The effects of confidence were examined in an analysis that excluded subjects in written conditions where confidence was not manipulated. Although the confidence effect was nonsignificant, $F(1, 160) = 1.45$, $p = .23$, the order of the means was consistent with the expectation that confidently as compared with nonconfidently presented messages would elicit greater opinion change ($M = 3.78$ vs. $M = 3.16$).

because, unlike our study, no measures of message comprehension were included. If their manipulation did not have a major impact on comprehension, then no effect on persuasion would be expected. Further, the fact that message complexity mattered in our study but not in the Miller et al. (1976) study may be due to the different experimental contexts employed. The conditions of our experiment, particularly its legal case context, probably maximized the tendency for recipients to focus attentively on the argumentation in the message and to base their opinions on such content cues. In such circumstances, because of heightened attention to message content, variables such as message complexity and communication modality may affect comprehension and, therefore, persuasion. Under other circumstances, however, recipients may not engage in such a cognitively elaborate process and may base their opinions on nonmessage cues such as communicator characteristics (e.g., Mills & Harvey, 1972; Norman, 1976). This is, in fact, the view favored by Miller et al. in interpreting the effects on persuasion of their speed of speech manipulation. In these circumstances, because of lack of attention to message content, variables such as message complexity and communication modality may have little effect on message comprehension and therefore little impact on persuasion.

Finally, because our design did not feature variation across stimulus conditions by including multiple topics or renditions of the difficulty manipulation, some concern about generality is in order. Yet, because variation across stimulus conditions was sacrificed in favor of charting the causal role of comprehension by developing a complex set of measures for a single set of stimulus materials, external validity concerns are of a particular sort. We would not argue that all easy and difficult communications would have effects across modalities of the kind we obtained. However, given the present results and the added generality they lend to the mediational role of comprehension in persuasion, we feel relatively confident in arguing that any communication difficult enough to create a comprehension deficit in the videotaped and audiotaped compared with written modality will be more persuasive if written. In the absence of such a comprehension deficit, we would expect greater persuasion in the videotaped compared with audiotaped and audiotaped compared with written modality.

REFERENCES

Baron, R. S., Baron, P. H., & Miller, N. The relation between distraction and persuasion. *Psychological Bulletin,* 1973, *80,* 310–323.

Beighley, K. C. An experimental study of the effect of four speech variables on listener comprehension. *Speech Monographs,* 1952, *19,* 249–258.

Brock, T. C. Communication discrepancy and intent to persuade as determinants of counterargument production. *Journal of Experimental Social Psychology,* 1967, *3,* 296–309.

Cantril, H., & Allport, G. W. *The psychology of radio.* New York: Harper Brothers, 1935.

Carver, M. E. Listening versus reading. In H. Cantril & G. W. Allport, *The psychology of radio.* New York: Harper Brothers, 1935.

Eagly, A. H. Comprehensibility of persuasive arguments as a determinant of opinion change. *Journal of Personality and Social Psychology,* 1974, *29,* 758–773.

Frandsen, K. D. Effects of threat appeals and media of transmission. *Speech Monographs,* 1963, *30,* 101–104.

Haaland, G. A., & Venkatesan, M. Resistance to persuasive communications: An examination of the distraction hypothesis. *Journal of Personality and Social Psychology,* 1968, *9,* 167–170.

Harwood, K. A. An experimental comparison of listening comprehensibility with reading comprehensibility. *Speech Monographs,* 1951, *18,* 123–124.

Haugh, O. M. The relative effectiveness of reading and listening to radio drama as ways of imparting information and shifting attitudes. *Journal of Educational Research,* 1952, *45,* 489–498.

Hovland, C. I., Janis, I. L., & Kelley, H. H. *Communication and persuasion.* New Haven, Conn.: Yale University Press, 1953.

Insko, C. A., Turnbull, W., & Yandell, B. Facilitative and inhibiting effects of distraction on attitude change. *Sociometry,* 1974, *37,* 508–528.

Janis, I. L., & Rife, D. Persuasibility and emotional disorder. In I. L. Janis & C. I. Hovland (Eds.), *Personality and persuasibility.* New Haven, Conn.: Yale University Press, 1959.

Keating, J. P., & Brock, T. C. Acceptance of persuasion and the inhibition of counterargumentation under various distraction tasks. *Journal of Experimental Social Psychology,* 1974, *10,* 301–309.

Knower, F. H. Experimental studies of changes in attitudes: I. A study of the effect of oral argument on changes of attitude. *Journal of Social Psychology,* 1935, *6,* 315–347.

Knower, F. H. Experimental studies of changes in attitudes: II. A study of the effect of printed argument on changes in attitude. *Journal of Abnormal and Social Psychology,* 1936, *30,* 522–532.

London, H. *Psychology of the persuader.* Morristown, N.J.: General Learning Press, 1973.

Maier, N. R., & Thurber, J. A. Accuracy of judgments of deception when an interview is watched, heard, and read. *Personal Psychology,* 1968, *21,* 23–30.

McGinnies, E. A cross-cultural comparison of printed communications versus spoken communications in persuasion. *Journal of Psychology,* 1965, *60,* 1–8.

McGuire, W. J. Order of presentation as a factor in "conditioning" persuasiveness. In C. I. Hovland (Ed.), *The order of presentation in persuasion.* New Haven, Conn.: Yale University Press, 1957.

McGuire, W. J. Personality and susceptibility to social influence. In E. F. Borgatta & W. W. Lambert (Eds.), *Handbook of personality theory and research.* Chicago: Rand McNally, 1968.

McGuire, W. J. The nature of attitudes and attitude change. In G. Lindzey & E. Aronson (Eds.), *The handbook of social psychology* (2nd ed., Vol. 3). Reading, Mass.: Addison-Wesley, 1969.

McGuire, W. J. Attitude change: The information-processing paradigm. In C. G. McClintock (Ed.), *Experimental social psychology.* New York: Holt, Rinehart & Winston, 1972.

Miller, N., Maruyama, G., Beaber, R. J., & Valone, K. Speed of speech and persuasion. *Journal of Personality and Social Psychology,* 1976, *34,* 615–624.

Mills, J., & Harvey, J. G. Opinion change as a function of when information about the communicator is received and whether he is attractive or expert. *Journal of Personality and Social Psychology,* 1972, *21,* 52–55.

Myers, J. L. *Fundamentals of experimental design* (2nd ed.). Boston: Allyn & Bacon, 1972.

Norman, R. When what is said is important: A comparison of expert and attractive sources. *Journal of Experimental Social Psychology,* 1976, *12,* 294–300.

Osterhouse, R. A., & Brock, T. C. Distraction increases yielding to propaganda by inhibiting counterarguing. *Journal of Personality and Social Psychology,* 1970, *15,* 344–358.

Tannenbaum, P. H., & Kerrick, J. S. Effect of newscast item leads upon listener interpretation. *Journalism Quarterly,* 1954, *31,* 33–37.

Westover, F. L. A comparison of listening and reading as a means of testing. *Journal of Educational Research,* 1958, *52,* 23–26.

Wilke, W. H. An experimental comparison of the speech, the radio, and the printed page as propaganda devices. *Archives of Psychology,* 1934, *25,* No. 169.

Williams, E. Medium or message: Communications medium as a determinant of interpersonal evaluation. *Sociometry,* 1975, *38,* 119–130.

Wilson, C. E. The effect of medium on loss of information. *Journalism Quarterly,* 1974, *51,* 111–115.

Worchel, S., Andreoli, V., & Eason, J. Is the medium the message? A study of the effects of media, communicator, and message characteristics on attitude change. *Journal of Applied Social Psychology,* 1975, *5,* 157–172.

Young, J. D. An experimental comparison of vocabulary growth by means of oral reading, silent reading, and listening. *Speech Monographs,* 1953, *20,* 273–276.

(Received December 12, 1975)